

Technical Architecture

The Ucab application follows a traditional Client-Server architecture utilizing the **MERN Stack** (MongoDB, Express, React, Node.js) with WebSockets for real-time bidirectional communication.

Key Components

- **Client Application:** A React SPA (Single Page Application) built with Vite, responsible for rendering the UI and maintaining user state.
- **REST API Server:** An Express.js application handling authentication, data processing, and business logic.
- **Database:** MongoDB NoSQL database to store dynamic JSON-like document data for users, drivers, and rides.
- **WebSocket Server:** A Socket.IO server integrated with Node.js to push live location updates and ride status changes instantly to clients.

Architecture Diagram

```
flowchart LR
    A["A[Client Browser / React]"] <-->|REST API (HTTP)| B["B(Express Node.js Server)"]
    A <-->|Socket.IO (WebSocket)| B
    B <-->|Mongoose ODM| C["C[(MongoDB Atlas)"]
```